

Alberto Brambila Solórzano, PhD

CFD, Thermal Systems & Simulation Automation Engineer

Mexico | Open to international relocation | English / Spanish

albertobraso7@gmail.com | +52 313 147 8404 | LinkedIn

Portfolio: albertobraso7.github.io

Professional profile

CFD engineer with experience across 30 professional projects in automotive thermal management, electric-machine cooling, hydrogen systems, valves and pumps. Develops physical models and reusable tools to improve design decisions and reduce manual simulation work. Combines multiphase and heat-transfer expertise with Python, OpenFOAM, PyFluent and 1D system-model calibration. Uses AI-assisted development and MCP-supported workflows with engineering review of APIs, assumptions and numerical results.

Professional experience

3D CFD Simulation Engineer | Bosch, Mexico

2022–present

- Developed an OpenFOAM-based hydraulic simulation GUI automating meshing, boundary conditions, convergence monitoring and results extraction. Reduced a standard four-case workflow from approximately 180 to 60 minutes; benchmarked pressure-drop predictions against Fluent. Most reported deviations were below 10%.
- Automated 1D valve calibration through Python and the Simcenter Amesim API, separating viscous and inertial regimes to fit physically identifiable parameters. The recorded calibrated model achieved RMSE no greater than 6.13% over -30 to 120°C against its 3D CFD reference.
- Created a repeatable preparation workflow for complex full-vehicle underhood geometry, reducing geometry preparation from weeks to two days. Delivered radiator airflow and thermal inputs for 1D system analysis and assessed operating and failure scenarios.
- Developed and applied a staged oil-cooling workflow within the team, combining shaft flow, VOF-to-DPM breakup, droplet impingement and conjugate heat transfer. Explained rotation-dependent cooling differences and provided heat-transfer maps for design evaluation.
- Built custom Python post-processing for transient pump pressure signals, including FFT, power spectral density and band-limited pressure RMS. Compared embedded and standalone arrangements beyond the commercial solver's standard reporting capabilities.
- Characterized compressible throttle flow across opening angles and pressure conditions for naturally aspirated and turbocharged applications; supported flow-bench comparison and delivery of operating curves.
- Investigated fuel-cell air cooling with spray injection, evaporation and wall-film models, using design comparisons to explain differences in cooling performance and guide a geometry recommendation.
- Evaluated pressure-loss, mixing and thermal-isolation changes in integrated coolant systems, translating local flow mechanisms into recommendations for mechanical design and system simulation teams.

Lecturer | Tecnológico de Colima, Mexico

2017–2022

Engineering teaching experience alongside graduate research, supporting clear explanation of technical methods and communication of analytical reasoning.

Simulation and programming capabilities

Physics: Multiphase VOF / Eulerian / DPM; sprays and wall films; conjugate heat transfer; turbulent and compressible flows; rotating machinery; species transport; porous media; fluid–structure interaction; non-Newtonian rheology.

Tools: Ansys Fluent, OpenFOAM, AVL FIRE M, Simcenter Amesim, PyFluent, Python, Bash, MATLAB, ANSA, ParaView, EnSight and Linux/HPC.

Methods: Parameter studies, design comparison, API-based solver control, automated post-processing, CPU/GPU benchmarking, numerical verification and CFD-to-test comparison. Workflow development includes settings verification, report dependencies and reusable configuration, with clear separation of setup, execution and analysis.

Selected engineering work and reusable methods

PyFluent automation and pump performance

Built configurable meshing, solver setup, operating-point sweeps and performance post-processing. In the documented generic pump benchmark, native GPU execution was 2.29 times faster than a 14-core CPU comparison, reducing wall time by 56.3% at the same 900 L/h operating point. Transient analysis separated startup from the spectral sampling window.

Generic demonstration: albertobraso7.github.io/pump/

Gas–liquid separation and setup reconstruction

Analyzed degassing mechanisms and geometry-dependent air transport. Developed reusable PyFluent mesh/setup workflows, checked boundary assignments and preserved report dependencies. The generic demonstration illustrates multiphase modeling and automation without disclosing an industrial component.

Generic demonstration: albertobraso7.github.io/debubbler/

Rapid screening, verification and model-selection judgment

Evaluated open-source meshing strategies and simplified geometry for hydraulic screening. Identified cases where geometric abstraction changed pressure-loss predictions materially. Assessed accelerated Eulerian and SPH oil-cooling methods, documenting sensitivity to tunable parameters before recommending their predictive use.

Hydrogen, sprays and thermal-management applications

Experience includes hydrogen dilution and ventilation, compressible injection, engine charge-motion and mixture studies, spray targeting, thermal-unit moisture/condensation, coolant-valve pressure loss and thermal isolation. Communicated findings to design, test and 1D teams.

Residential airflow and solar assessment

Prepared building-flow models, compared baseline and modified layouts, and used three-dimensional streamlines and vertical sections to explain room-level circulation. Assessed solar exposure separately from airflow, preserving the different conditions and interpretation limits.

Consulting case: albertobraso7.github.io/ventilation/

Research and education

PhD, Thermofluids Engineering | UNAM

2025

Studied aneurysm hemodynamics using OpenFOAM, including blood rheology, turbulence-model assessment, fluid–structure interaction and particle transport. Used benchmark comparisons and numerical sensitivity checks to distinguish physical-model effects from numerical artifacts. The research demonstrates coupled-flow methods, not clinical prediction.

Research case: albertobraso7.github.io/aneurysm/

Master of Engineering | Universidad de Colima

Developed CFD and experimental analysis of a tubular CVD reactor. Used Reynolds, Grashof, Richardson and Nusselt reasoning to interpret buoyancy, porous flow and precursor placement, connecting thermal modeling with experimental observations.

Research case: albertobraso7.github.io/reactor/

Mechanical and Electrical Engineering | Universidad de Colima

2014

Selected publications

Brambila-Solórzano, A., et al. Influence of blood rheology and turbulence models in the numerical simulation of aneurysms. *Bioengineering* 10, 1170 (2023).

Sánchez, G. M., et al. Numerical simulation of the aqueous humor flow in the eye drainage system; a healthy and pathological condition comparison. *Medical Engineering & Physics* 83, 82–92 (2020).

Portfolio context

Public industrial examples use simplified or alternative geometries to illustrate methods and skills applied in professional work. They do not reproduce customer components or disclose project-specific results. Academic studies are identified separately. Implementation files remain private.